



MILESTONE 7

Harden It: CI/CD and Reliability

Treating the pipeline like production code: version control, automated tests, and run auditing



STEP 1

Everything in Version Control

WHAT

Put the whole project in one Git repository: OLTP SQL, the ADF pipeline JSON, the Databricks notebooks, the dbt project, and the Power BI report, each pushed from its own tool. ADF and Databricks are Git-connected; a top-level README documents the structure.

WHY

A project a business can trust is versioned, reviewable, and reproducible from the repo alone. Pushing each artifact from its own tool keeps the committed code identical to what actually runs.

Everything in Version Control

The screenshot displays the GitHub web interface for the repository 'atliq-commerce-capstone' by user 'ashishbabaria'. The left sidebar shows the file tree with the 'phase1-batch' directory selected. The main content area shows the commit history for 'phase1-batch', with the most recent commit by 'ashishbabaria' (M7: root README, remove orphaned ingestion_adf, add run_log verify query) highlighted. Below the commit history, the README file is open, showing the title 'AtliQ Commerce Data Engineering Capstone' and a section titled 'Repository structure' which includes a tree diagram of the project layout.

Files

main

Go to file

.github/workflows

ci.yml

phase1-batch

adf

databricks

dbt/atliq_gold

fabric

oltp

.gitignore

README.md

README.md

atliq-commerce-capstone / phase1-batch

ashishbabaria M7: root README, remove orphaned ingestion_adf, add run_log verify query 910bae9 · 5 hours ago History

Name	Last commit message	Last commit date
..		
adf	Updating pipeline: pl_nightly_sync	5 hours ago
databricks	Add run_dbt notebook (M5 orchestration)	last week
dbt/atliq_gold	Add CI workflow and env-var profiles (M7)	9 hours ago
fabric	Add M6 Power BI dashboard (.pbix)	last week
oltp	M7: root README, remove orphaned ingestion_adf, add run_log verify query	5 hours ago
.gitignore	Create .gitignore for project dependencies and artifacts	28 days ago
README.md	Initial commit: Phase 1 OLTP scripts, README	28 days ago

README.md

AtliQ Commerce Data Engineering Capstone

End-to-end data engineering for AtliQ Commerce, an online storefront, built as two independent lanes that run side by side, the way batch and streaming coexist in real companies.

Repository structure

```
atliq-commerce-capstone/  
├── phase1-batch/      Batch lane: nightly OLTP to lakehouse to Fabric  
└── phase2-streaming/  Speed lane: real-time order events (added in Phase 2)
```

Phase 1: Batch Lane

GitHub repo atliq-commerce-capstone: phase1-batch with adf, databricks, dbt, fabric, oltp and the CI workflow

Everything in Version Control

The screenshot displays the Microsoft Azure Data Factory portal interface. The top navigation bar shows 'Microsoft Azure | Data Factory | adf-atliq-ab' with a search bar and user profile 'ashish@alphainterbiz.com'. A green banner at the top right encourages upgrading to the 'Microsoft Fabric Data Factory' experience, with a 'View in Fabric (Preview)' button.

The left sidebar contains a navigation menu with categories: General, Connections, Source control, ARM template, Author, Security, and Workflow orchestration manager. The 'Source control' category is expanded, showing 'Git configuration' as the selected option.

The main content area is titled 'Git repository' and provides information about the associated Git repository. It includes a table of configuration details:

Property	Value
Repository type	GitHub
GitHub account	ashishbabaria
Repository name	atliq-commerce-capstone
Collaboration branch	main
Publish branch	adf_publish
Root folder	/phase1-batch/adf
Last published commit	b7490c3198ab9f35780a3a0dd777dde0b6dc845d
Publish (from ADF Studio)	Enabled
Custom comment	Disabled

ADF Git configuration: GitHub repo, collaboration branch main, root folder /phase1-batch/adf

Everything in Version Control

The screenshot displays the GitHub web interface for the repository 'atliq-commerce-capstone'. The left sidebar shows the file structure, with 'phase1-batch/adf' selected. The main area shows the commit history for the 'adf' directory, listing files like 'dataset', 'factory', 'linkedService', 'pipeline', 'trigger', and 'publish_config.json' with their commit messages and dates.

Files in the repository:

- phase1-batch
 - adf
 - dataset
 - ds_adls_bronze_parquet.json
 - ds_adls_csv_marketing.json
 - ds_adls_csv_supplier.json
 - ds_azuresql_control.json
 - ds_azuresql_generic.json
 - factory
 - adf-atliq-ab.json
 - linkedService
 - ls_adls.json
 - ls_azuresql.json
 - ls_databricks.json
 - ls_keyvault.json
 - pipeline
 - pl_nightly_sync.json
 - trigger
 - trg_nightly.json
 - publish_config.json
 - databricks
 - dbt/atliq_gold
 - fabric
 - oltp

Commit history for 'adf':

Name	Last commit message	Last commit date
..		
dataset	Adding dataset: ds_azuresql_generic	28 days ago
factory	Adding dataset: ds_azuresql_generic	28 days ago
linkedService	Adding linkedService: ls_databricks	last week
pipeline	Updating pipeline: pl_nightly_sync	5 hours ago
trigger	Updating trigger: trg_nightly	last week
publish_config.json	Create publish_config.json	28 days ago

phase1-batch/adf: pipeline, datasets, linked services and trigger JSON, committed by ADF itself

1

STEP 1 · SCREENSHOT 4 of 6

Everything in Version Control

The screenshot shows the GitHub interface for the repository 'atliq-commerce-capstone' by user 'ashishbabaria'. The left sidebar displays the file structure, with the 'databricks' folder selected under 'phase1-batch'. The main area shows the commit history for the 'databricks' folder, listing five commits with their names, messages, and dates.

Repository: atliq-commerce-capstone

Branch: main

Files:

- .github/workflows
- ci.yml
- phase1-batch
 - adf
 - databricks
 - 00_uc_setup.ipynb
 - 10_silver_full_dims.ipynb
 - 20_silver_incremental_merge.i...
 - run_dbt.ipynb
 - dbt/atliq_gold
 - fabric
 - oltp
 - .gitignore
 - README.md
 - README.md

Commit History:

Name	Last commit message	Last commit date
..		
00_uc_setup.ipynb	Add M3 Silver notebooks and Unity Catalog setup	28 days ago
10_silver_full_dims.ipynb	Add M3 Silver notebooks and Unity Catalog setup	28 days ago
20_silver_incremental_merge.ipynb	Add M3 Silver notebooks and Unity Catalog setup	28 days ago
run_dbt.ipynb	Add run_dbt notebook (M5 orchestration)	last week

phase1-batch/databricks: the Silver notebooks and run_dbt, pushed from the Databricks Git folder

1

STEP 1 · SCREENSHOT 5 of 6

Everything in Version Control

The screenshot shows the GitHub interface for the repository 'atliq-commerce-capstone' by user 'ashishbabaria'. The left sidebar displays the file tree for the 'main' branch, with the 'dbt/atliq_gold' directory selected. The main area shows the commit history for the selected path, listing files and their last commit messages and dates. Below the table, the 'README.md' file content is visible, providing instructions for setting up a new dbt project.

Files

main

Go to file

dbt/atliq_gold

- analyses
- macros
- models
 - gold
 - dim_customer.sql
 - dim_date.sql
 - dim_product.sql
 - fact_sales.sql
 - schema.yml
 - staging
 - sources.yml
 - stg_customers.sql
 - stg_order_items.sql
 - stg_orders.sql
 - stg_payments.sql
 - stg_products.sql
 - stg_supplier_price_list.sql
 - seeds
 - snapshots
 - tests
- .gitignore
- .user.yml
- README.md
- dbt_project.yml
- packages.yml
- profiles.yml

atliq-commerce-capstone / phase1-batch / dbt / atliq_gold

ashishbabaria Add CI workflow and env-var profiles (M7) 844d498 · 9 hours ago History

Name	Last commit message	Last commit date
..		
analyses	Add M4 dbt project (Gold star schema)	28 days ago
macros	Add M4 dbt project (Gold star schema)	28 days ago
models	Add M4 dbt project (Gold star schema)	28 days ago
seeds	Add M4 dbt project (Gold star schema)	28 days ago
snapshots	Add M4 dbt project (Gold star schema)	28 days ago
tests	Add M4 dbt project (Gold star schema)	28 days ago
.gitignore	Add M4 dbt project (Gold star schema)	28 days ago
.user.yml	Add run_dbt notebook (M5 orchestration)	last week
README.md	Add M4 dbt project (Gold star schema)	28 days ago
dbt_project.yml	Add M4 dbt project (Gold star schema)	28 days ago
packages.yml	Add M4 dbt project (Gold star schema)	28 days ago
profiles.yml	Add CI workflow and env-var profiles (M7)	9 hours ago

README.md

Welcome to your new dbt project!

Using the starter project

Try running the following commands:

phase1-batch/dbt/atliq_gold: the dbt project, models, tests and env-var profiles.yml

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STEP 1 · SCREENSHOT 6 of 6

Everything in Version Control

The screenshot shows the GitHub web interface for the repository 'atliq-commerce-capstone' by user 'ashishbabaria'. The top navigation bar includes links for Code, Issues, Pull requests, Agents, Actions, Projects, Wiki, Security and quality, Insights, and Settings. A search bar is located on the right. The left sidebar displays the file explorer with the 'main' branch selected. The file structure is as follows:

- .github/workflows
- phase1-batch
 - adf
 - databricks
 - dbt/atliq_gold
 - fabric**
 - atliq-sales-dashboard.pbix
 - oltp
 - csv
 - 01_schema_ddl.sql
 - 02_insert_customers.sql
 - 03_insert_products.sql
 - 04_insert_orders.sql
 - 05_insert_order_items.sql
 - 06_insert_payments.sql
 - 07_etl_control_table.sql
 - 08_audit_run_log.sql
 - README.md
 - daily_order_simulator.py
 - verify_counts.sql
 - verify_run_log.sql
 - .gitignore
 - README.md
 - README.md

The main content area shows the commit history for the 'fabric' directory. The commit message is 'Add M6 Power BI dashboard (.pbix)' by 'ashishbabaria' (7bb2f88) from last week. Below this, a table lists the files in the commit:

Name	Last commit message	Last commit date
..		
atliq-sales-dashboard.pbix	Add M6 Power BI dashboard (.pbix)	last week

phase1-batch/fabric (Power BI .pbix) and oltp (SQL scripts 01 to 08, verify queries)



STEP 2

Secrets Out of Code

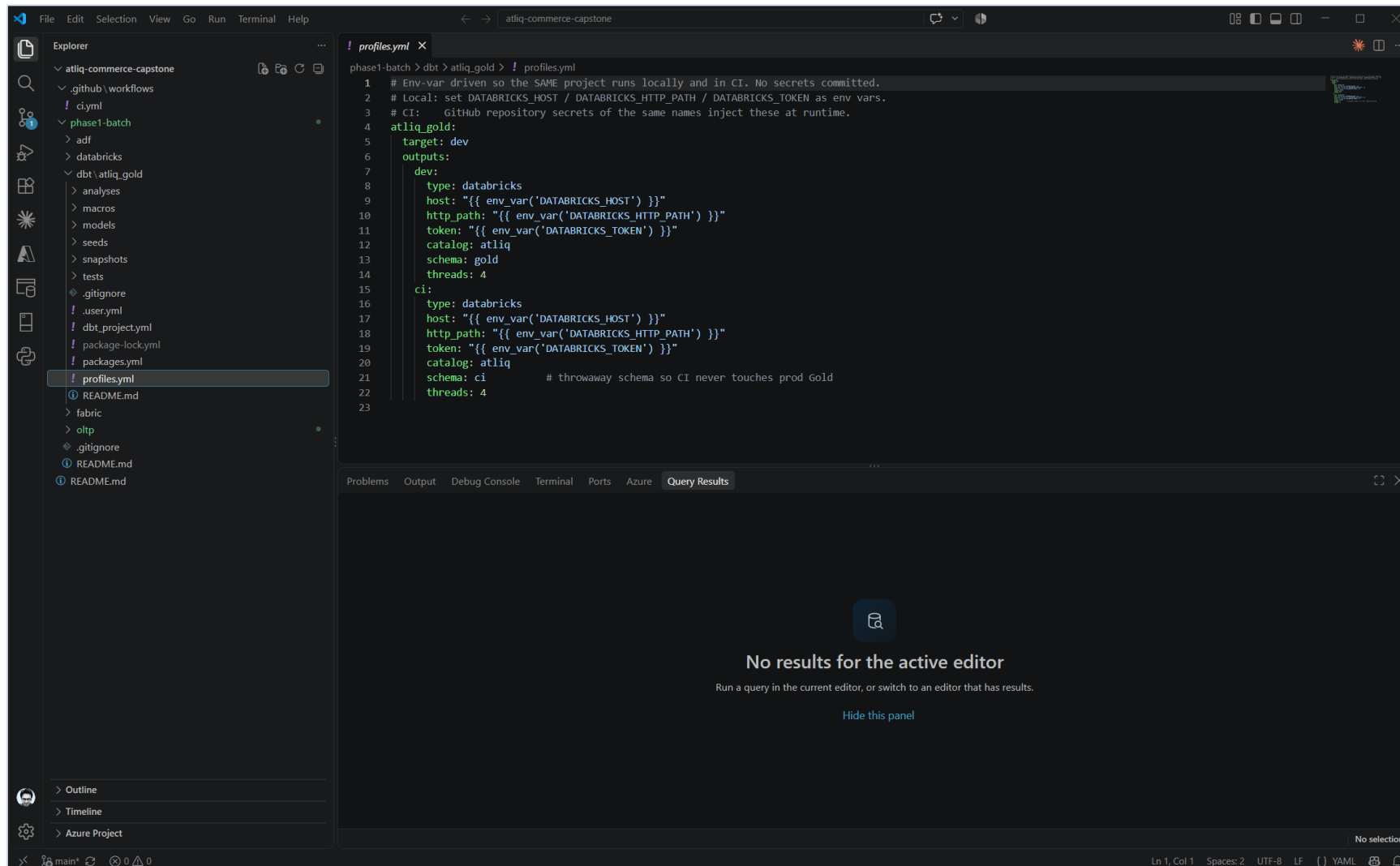
WHAT

dbt reads its connection from environment variables, so the same project runs locally and in CI with no credentials committed. A separate ci target builds into a throwaway ci schema, never touching production Gold.

WHY

Committing a token is a leak waiting to happen. Env-var configuration plus a throwaway CI schema means the same code is safe to run anywhere, and CI can never corrupt the real Gold tables.

Secrets Out of Code



profiles.yml: host, http_path and token read via env_var, with dev (gold) and ci (throwaway schema) targets

Secrets Out of Code

The screenshot shows the GitHub Actions repository settings page for the repository 'atliq-commerce-capstone' by user 'ashishbabaria'. The 'Settings' tab is selected in the top navigation bar. On the left sidebar, the 'Secrets and variables' section is expanded, showing options like General, Access, Collaborators, Moderation, Code, planning, and automation, Security and quality, Integrations, and GitHub Apps. The main content area is titled 'Actions secrets and variables'. It contains a description of secrets and variables, a section for 'Environment secrets' (currently empty with a 'Manage environment secrets' button), and a section for 'Repository secrets'. The 'Repository secrets' section has a 'New repository secret' button and a table listing three secrets: DATABRICKS_HOST, DATABRICKS_HTTP_PATH, and DATABRICKS_TOKEN, each with its last updated time and edit/delete icons.

ashishbabaria / atliq-commerce-capstone

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

General

Access

Collaborators

Moderation

Code, planning, and automation

Rulesets

Branches

Tags

Actions

Webhooks

Copilot

Planning

Environments

Codespaces

Pages

Security and quality

Advanced Security

Deploy keys

Secrets and variables

Integrations

GitHub Apps

Email notifications

Actions secrets and variables

Secrets and variables allow you to manage reusable configuration data. Secrets are **encrypted** and are used for sensitive data. [Learn more about encrypted secrets](#). Variables are shown as plain text and are used for **non-sensitive** data. [Learn more about variables](#).

Anyone with collaborator access to this repository can use these secrets and variables for actions. They are not passed to workflows that are triggered by a pull request from a fork.

Secrets Variables

Environment secrets

This environment has no secrets.

Manage environment secrets

Repository secrets

New repository secret

Name ↕	Last updated
DATABRICKS_HOST	3 minutes ago
DATABRICKS_HTTP_PATH	3 minutes ago
DATABRICKS_TOKEN	1 minute ago

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GitHub Actions repository secrets: DATABRICKS_HOST, DATABRICKS_HTTP_PATH, DATABRICKS_TOKEN



STEP 3

Continuous Integration

WHAT

Added a GitHub Actions workflow that runs on every pull request: it installs dbt, then runs dbt build against the ci target, building all models and running all sixteen tests, with the Databricks connection injected from repository secrets.

WHY

CI turns the test suite into a gate. A pull request that would break a model or violate a data-quality rule fails visibly before it can merge, so main always holds a project that builds and passes.

Continuous Integration

Add root README and trigger CI #1

Merged

ashishbabaria merged 2 commits into `main` from `test-ci` now

Conversation 0Commits 2Checks 1Files changed 2+3 -2

ashishbabaria commented 1 hour ago

No description provided.

ashishbabaria added 2 commits 1 hour ago

Add root README and trigger CI

Fix dbt profiles-dir in CI workflow

ashishbabaria merged commit a1540dd into main now

1 of 2 checks passed

View details

Revert

Pull request successfully merged and closed

You're all set — the `test-ci` branch can be safely deleted.

Delete branch

Add a comment

Write

Preview

H B I

Add your comment here...

Markdown is supported

Paste, drop, or click to add files

Comment

Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

ProTip! Add `.patch` or `.diff` to the end of URLs for Git's plaintext views.

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

Customize

Unsubscribe

You're receiving notifications because you modified the open/close state.

1 participant

Lock conversation

Archive pull request

The fix: failed commit, then the profiles-dir fix passing, then the PR merged into main

Continuous Integration

The screenshot shows the GitHub Actions interface for a workflow named "Add root README and trigger CI #2". The workflow is triggered by a pull request. The run status is "Success" with a total duration of 1m 37s. The workflow file is named "ci.yml" and is located on the "pull_request" event. The workflow contains a single job named "build" which has a status of "Success" and a duration of 1m 32s. The workflow file content is as follows:

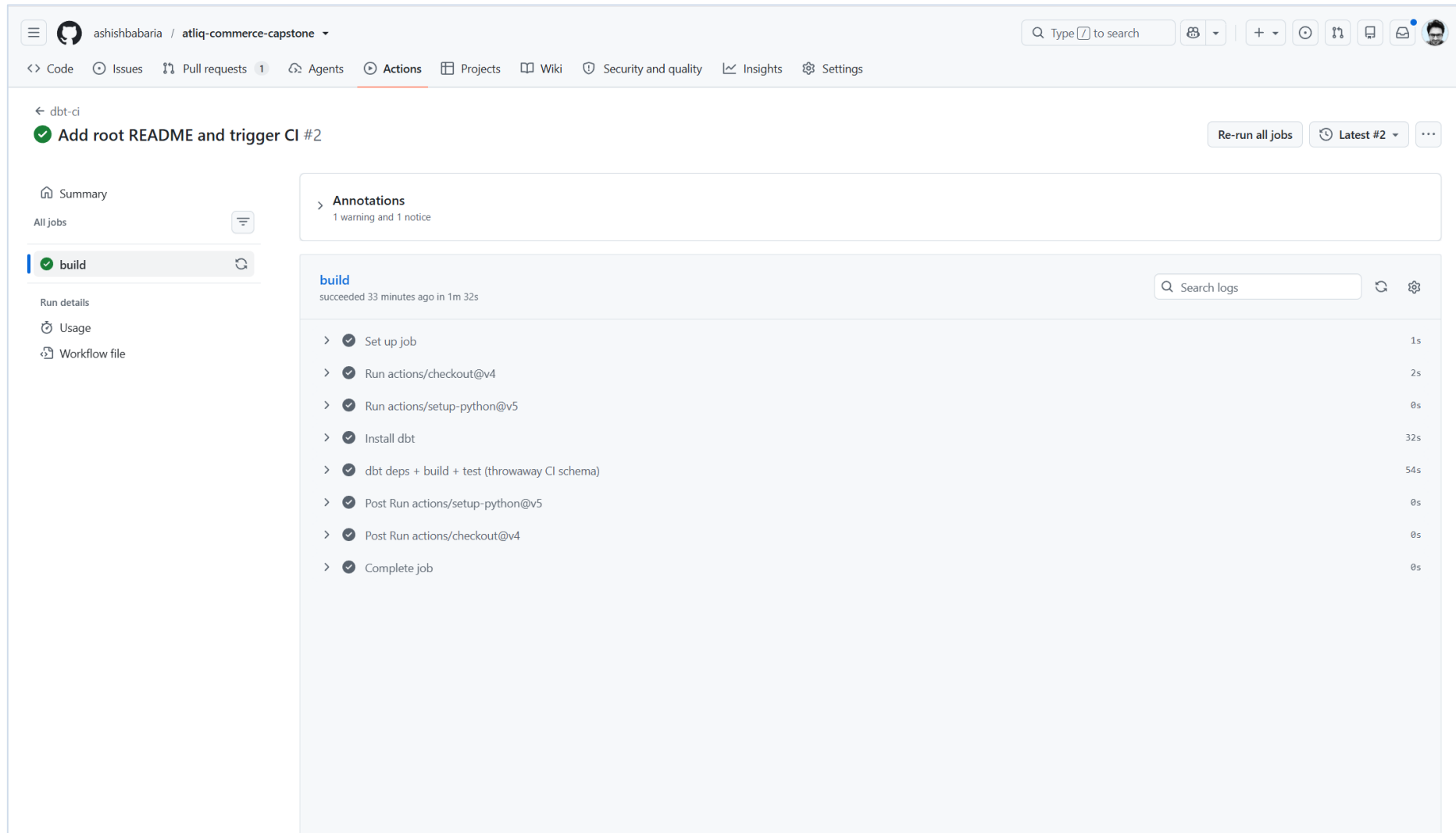
```
ci.yml
on: pull_request

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
      - run: pip install -r requirements.txt
      - run: python manage.py test
```

The "Annotations" section shows 1 warning and 1 notice. The warning is for the "build" job, stating that Node.js 20 is deprecated and the following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setup-python@v5. The notice is for the "build" job, stating that the ubuntu-latest label will migrate to Ubuntu 26 beginning October 19, 2026.

ci.yml on pull_request: build job Success

Continuous Integration



The screenshot displays the GitHub Actions interface for a workflow named 'dbt-ci'. The job 'Add root README and trigger CI #2' is shown as successful. The left sidebar contains navigation links: Summary, All jobs, build (selected), Run details, Usage, and Workflow file. The main area shows the job's annotations, indicating 1 warning and 1 notice. Below this, the 'build' job is detailed, showing it succeeded 33 minutes ago in 1m 32s. The job steps are listed with their durations:

Step	Duration
Set up job	1s
Run actions/checkout@v4	2s
Run actions/setup-python@v5	0s
Install dbt	32s
dbt deps + build + test (throwaway CI schema)	54s
Post Run actions/setup-python@v5	0s
Post Run actions/checkout@v4	0s
Complete job	0s

CI build job steps, including dbt deps + build + test on the throwaway CI schema



STEP 4

Failure Alerting

WHAT

Configured an Azure Monitor alert on the ADF pipeline that emails on any failed run, so a broken nightly job notifies instead of silently serving stale data.

WHY

A silently failing pipeline is worse than a visibly down one: the dashboard keeps showing yesterday's numbers as if they were today's. An alert makes failure loud.

Failure Alerting

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

ashish@alphainterbiz.co...
ALPHA INTERBIZ LLP (ALPHAINT...)

Home > Monitor | Alerts > Action groups > atliq-alerts

atliq-alerts

Edit action group

Save changes

Basics

Subscription: Azure subscription 1

Resource group: rg-atliq-capstone

Region: eastus

Action group name: atliq-alerts

Display name *: AtliqAlert ✓

Notifications

Notification type	Name	Status	Selected
Email/SMS message...	Email	Subscribed	Email

Actions

Action type	Name	Selected	Identity

Tags

Tags are name and value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about using tags](#)

Name ⓘ Value ⓘ

:

Email/SMS message/Push/Voice

Add or edit Email/SMS message/Push/Voice action

☒ Email

Email * ⓘ

Please note: Emails must be verified before the action group can send notifications. Unverified emails will receive a verification email.

☐ SMS (Carrier charges may apply)

Country code: 1

Phone number:

☐ Azure mobile app notification

Azure account email ⓘ

☐ Voice

Country code: 1

Phone number:

Enable the common alert schema. [Learn more](#)

Yes No

OK

Action group atliq-alerts: email notification subscribed, used by the ADF failed-run alert (address blurred)



STEP 5

Run Auditing

WHAT

Wired the pipeline to an audit table (etl.run_log): a stored procedure logs a started row at the beginning of each run, and a second marks it success with the end time when the run completes. A verify query reads the log.

WHY

The audit table is the pipeline's black box recorder. If a dashboard ever looks wrong, this is the first place to check: what ran, when, and whether it passed. It is what makes the system trustworthy, not just functional.

Run Auditing

Microsoft Azure | Data Factory | adf-atliq-ab

Upgrade your Azure Data Factory to the native **Microsoft Fabric Data Factory** experience. [View in Fabric \(Preview\)](#)

main branch | Validate all | Save all | Publish

Activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

pl_nightly_sync

Activities

- Set variable (X) Set run_start_at
- Stored procedure Log run start
- Lookup Lookup control table
- ForEach ForEach source
- Copy data Copy supplier CSV
- Copy data Copy marketing CSV
- Notebook Silver full dims
- Notebook Silver incremental
- Notebook Run dbt

Parameters Variables Settings **Output**

Pipeline run ID 00e76efa-91b6-4372-8ef2-f8b5d4ecc8bf Pipeline status **Succeeded** [View debug run consumption](#)

All status List

Showing 1 - 23 of 23 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID	Log
Log run success	✓ Succeeded	Stored procedu	9/29/2026, 1:52:06 PM	16s	AutoResolveIntegrationRuntime (Central India)		4122da6f-12b1-411e-8826-3b0b041b0f04	
Run dbt	✓ Succeeded	Notebook	9/29/2026, 1:47:39 PM	4m 25s	AutoResolveIntegrationRuntime (Central India)		bc56d820-6375-4e37-a576-2f0cfd65f3af	
Silver incremental	✓ Succeeded	Notebook	9/29/2026, 1:43:51 PM	3m 42s	AutoResolveIntegrationRuntime (Central India)		138a4730-f3f7-4d41-8b55-b3af0a04b9f6	
Silver full dims	✓ Succeeded	Notebook	9/29/2026, 1:37:07 PM	6m 41s	AutoResolveIntegrationRuntime (Central India)		bc815892-f8ff-4969-a6a3-16815b57c4ff	
Advance watermark	✓ Succeeded	Stored procedu	9/29/2026, 1:36:41 PM	6s	AutoResolveIntegrationRuntime (Central India)		df17bb2d-6881-4a32-8152-0fa6933760f9	
Copy full	✓ Succeeded	Copy data	9/29/2026, 1:36:39 PM	21s	AutoResolveIntegrationRuntime (Central India)		56e559d2-614e-4b01-976a-1e960abb771	
Advance watermark	✓ Succeeded	Stored procedu	9/29/2026, 1:36:38 PM	6s	AutoResolveIntegrationRuntime (Central India)		449df9fc-db04-41a4-bee9-cc0cb4b28112	

pl_nightly_sync with Log run start and Log run success wrapped around the chain, run Succeeded

Run Auditing

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'etl.run_log' table under the 'etl.run_log' schema. The right pane shows the query results for the 'verify_run_log.sql' query. The query is as follows:

```
1 -- Verify the pipeline audit log: shows each nightly run's
2 -- start time, end time, stage, status, and message.
3 SELECT run_id, run_start_at, run_end_at, stage, object_name, rows_affected, status, message
4 FROM etl.run_log
5 ORDER BY run_id DESC;
```

The query results are displayed in the 'Query Results' pane. The results show a single row with the following data:

run_id	run_start_at	run_end_at	stage	object_name	rows_affected	status	message
3	2026-09-29 08:05:42	2026-09-29 08:22:19	nightly	NULL	NULL	success	Nightly pipeline completed

The status bar at the bottom indicates 'Rows: 1', 'Time: 20ms', and 'No selection'.

etl.run_log: nightly run logged with start, end and status success

Milestone 7 Complete

What this milestone delivered

Versioned and reproducible

The whole project in one Git repo, each artifact pushed from its own tool, with secrets kept out of code via environment variables.

Tested on every change

GitHub Actions runs dbt build and all sixteen tests on each pull request, into a throwaway CI schema that never touches production Gold.

Trustworthy in production

A failure alert makes broken runs loud, and an audit table records every run's start, end, and status.

Reliability at a glance

1

Git repo, all artifacts

16

tests gate every PR

1

failure alert

1

audit log per run

Phase 1 complete: OLTP to lakehouse to Fabric, orchestrated as one scheduled idempotent nightly job, reported through Power BI, and hardened with CI, alerting, and run auditing. The batch lane is a system a business can trust, not a one-off script.